for understanding

HAMMING CODE

* Given by R.W. Hamming
* Easy to implement
* 7- bit hamming code is used commonly. can have more than 7 bit

Data bits

Parity bits

↳ an extra stream of bits which we send along with data bit to detect error.

In 7 bit hamming code

3 - bits → Parity bit

4 - bits → Data bit.

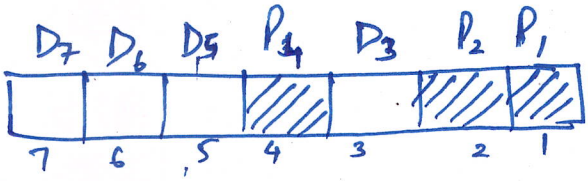
2^n { where $n = 0, 1 \cdots n$ }

↳ Position for parity bits

$2^0 \rightarrow 1$

$2^1 \rightarrow 2$

$2^2 \rightarrow 4$

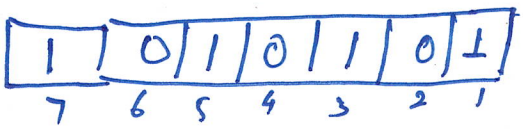


$P_1 \rightarrow D_3\ D_5\ D_7$

$P_2 \rightarrow D_3\ D_6\ D_7$

$P_4 \rightarrow D_5\ D_6\ D_7$

Ex. Let suppose, we want to transmit 4-bit data (1011)



T_x ——— 1010101 ——— R_x

$P_1 = 1\ 1\ 1$

↳ 1 for even parity.

$P_2 \rightarrow 1\ 0\ 1$

↳ 0

$P_4 \rightarrow 1\ 0\ 1$

Now, let suppose the transmitted bit stream that we have received is now

1 (1) 1 0 1 0 1

Receiver will check for parity

$P_1 = 1$

$P_2 = 0$

$P_4 = 0$

$D_3\ D_6\ D_7$
1 1 1
$D_5\ D_6\ D_7$
1 1 1
↳ error again.

there is an error as we are checking for even-parity

Now, we know that there is error in either D_3, D_5, D_6 or D_7.

Q. If a 7-bit hamming code received by a receiver is 1 0 1 1 0 1 1. Assuming the even parity. State whether the received code word is correct or wrong. If wrong locate the bit having error.

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	1	0	1	1

$P_4 = 1$ $D_5\ D_6\ D_7$
1 0 1
↳ error. (Parity bit as)

Whenever there is an error but $P_4 = 1$ here.

$P_2\ D_3\ D_6\ D_7$
1 0 0 1 → even.

$\boxed{P_2 = 0}$ (as no contradiction)

$P_1\ D_3\ D_5\ D_7$
1 0 1 1 → error.

$\boxed{P_1 = 1}$

$P_4\ P_2\ P_1$

(so the 5th bit is having error. thus $D_5 = 0$.

Ans 1 0 0 1 0 1 1
↓
correct code

Detecting & Correcting Error

* Each parity bit, along with its corresponding group of bits must be checked for proper parity.
* The correct result of individual parity is marked by 0 whereas wrong result is marked by 1.
* After all parity checks, binary word is formed taking resulting bit for P_1 as LSB. This word gives bit location, where error has occured.

Q. Assume that the even parity Hamming code is transmitted and 0100011 is received. The receiver code does not know what was transmitted. Determine bit location where error has occured using received code.

D_7	D_6	D_5	P_4	D_3	P_2	P_1
7	6	5	4	3	2	1
111	110	101	100	011	010	001
0	1	0	0	0	1	1

For P_1 P_1 checks location 1, 3, 5, 7.

Parity is wrong — 1

P_2 checks location 2, 3, 6 and 7

Parity is correct → 0

P_4 checks location 4, 5, 6 and 7

Parity is wrong → 1

Resultant $(101)_2 \rightarrow 5_{10}$ The bit location 5 has error. It is 0 & should be 1.

Correct code :- 0110011.

Q. The Hamming code 101101101 is received. Correct it if any errors. There are four parity bits and odd parity is used.

D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1
9	8	7	6	5	4	3	2	1
1001	1000	0111	0110	0101	0100	0011	0010	001
1	0	1	1	0	1	1	0	1

$P_1 \rightarrow$ 1, 3, 5, 7 and 9.

P_1 is wrong 1.

$P_2 \rightarrow$ 2, 3, 6 and 7

$P_2 \rightarrow 0$

$P_4 \rightarrow$ 4, 5, 6, and 7

$P_4 \rightarrow 0$

$P_8 \rightarrow$ 8 and 9.

$P_8 \rightarrow 0$

Parity :- $(0001)_2$

$= (1)_{10}$

The correct code :- 101101100.

Q. Detect and correct error (if any) in the following received even parity hamming code word 001111101010.

Also find the correct msg.

find the no. of parity bits

$$2^P \geq x + P + 1$$

$$2^P \geq 12 + P$$

$$2^P \geq 12$$

let $P = 4$ $\quad 2^4 \geq 12 + 4$

Thus $P = 4$

D_{11}	D_{10}	D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1
11	10	9	8	7	6	5	4	3	2	1
1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001
0	0	1	1	1	1	0	1	0	1	0

$P_1 \rightarrow$ 1, 3, 5, 7, 9 and 11

$P_1 \rightarrow 0$

$P_2 \rightarrow$ 2, 3, 6, 7, 10 & 11

$P_2 \rightarrow 1$

$P_4 \rightarrow$ 4, 5, 6 & 7

$P_4 \rightarrow 1$

$P_8 \rightarrow$ 8, 9, 10, 11

$P_8 \rightarrow 0$.

Resultant word $(0110)_2 \rightarrow (6)_{10}$

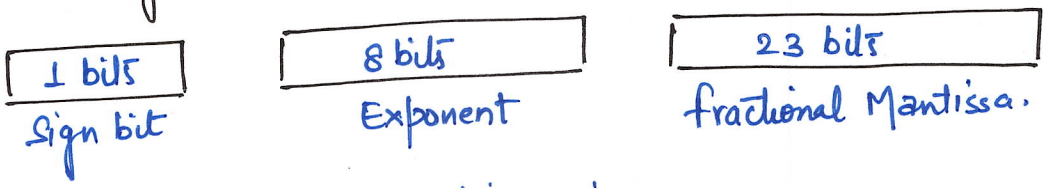
6 location has error.

$1 \rightarrow 0$

Correct msg $\rightarrow$ 00111001010

FLOATING POINT REPRESENTATION

(a) 32-bit (Single Precision).



* Sign bit represents the mantissa sign.
  0 for positive & 1 for negative.
* Exponent is stored in biased 127, i.e, (exponent +127)
* fractional Mantissa stores the no. after decimal point.

Example:- Represent 11.101×2^3 in IEEE 32-bit floating point format.

$11.101 \times 2^3 = 1.1101 \times 2^4$

(Normalized form)

Exponent = (127 + 4)

$= (131)_{10} = (10000011)_2$

Ans:- 0 | 10000011 | 11010.....0

1 bit | 8 bit | 23 bit.

Ex Represent -0.001011×2^2 in IEEE 754 32-bit floating point representation.

$-0.001011 \times 2^2 = -1.011 \times 2^{-1}$

(Normalized form)

$(127 - 1) = (126)_{10}$

$= (01111110)_2$

1 | 01111110 | 01100.....0

1 bit | 8 bit | 23-bit

64-bit (Double Precision)

1 bit	11 bits	52 bits.
Sign bit	Exponent (1023 biased)	fractional Mantissa.

Q. Represent $(-15.25)_{10}$ in IEEE 754 64-bit floating point representation.

$(-15.25)_{10} = (-1111.01)_2$

$= -1.11101 \times 2^3$

(Normalized form)

$(1023 + 3) = (1026)_{10}$

$= (10000000010)_2$

BOOLEAN ALGEBRA

Boolean algebra is a mathematical system that defines a series of logical operations (AND, OR, NOT) performed on sets of variables (a, b--c) when stated in this form, the expression is called a

Boolean Equation

Variable :- The symbol which represent an arbitrary elements of an Boolean Algebra is known as variable

Constant :- The value is fixed $Y = A + 1$, A is variable, 1 is constant

Complement :- The complement of A, is $\bar{A}$

Literal :- Each occurance of a variable in Boolean function either in a complemented or uncomplemented form is called a literal

Boolean Theorems & Postulates

Postulates

Identity	$A+0 = A$	$A \cdot 1 = A$
Commutative	$A+B = B+A$	$AB = BA$
Distributive	$A(B+C) = AB+AC$	$A+BC = (A+B)(A+C)$
Complement	$A+\bar{A} = 1$	$A \cdot \bar{A} = 0$

Theorems

Idempotency	$A+A = A$	$A \cdot A = A$
Theorem-2	$A+1 = 1$	$A \cdot 0 = 0$
Involution	$\bar{\bar{A}} = A$	
Absorption	$A+AB = A$	$A(A+B) = A$
Theorem 5	$A+\bar{A}B = A+B$	$A \cdot (\bar{A}+B) = AB$
Associative	$A+(B+C) = (A+B)+C$	$A(BC) = (AB)C$

Q. Simplify $\bar{A}BC\bar{D} + BC\bar{D} + B\bar{C}\bar{D} + B\bar{C}D$

$= BC\bar{D}(\bar{A}+1) + B\bar{C}\bar{D} + B\bar{C}D$

$= BC\bar{D} + B\bar{C}\bar{D} + B\bar{C}D$

$= B\bar{D}(C+\bar{C}) + B\bar{C}D$

$= B\bar{D} + B\bar{C}D$

$= B(\bar{D} + \bar{C}D) \quad \longrightarrow A + \bar{A}B = A + B.$

$= B(\bar{D}+\bar{C})$

Q. Simplify $AB + \bar{A}\bar{C} + A\bar{B}C(AB+C)$

$= AB + \bar{A}\bar{C} + AA\bar{B}\overbrace{B\bar{C}}^{=0} + A\bar{B}CC$

$= AB + \bar{A}\bar{C} + A\bar{B}CC$

$= AB + \bar{A}\bar{C} + A\bar{B}C$

$= \underline{AB + \bar{A}} + \bar{C} + A\bar{B}C \quad \longrightarrow A + \bar{A}B = A + B$

$= \cancel{AB}\ \bar{A} + B + \bar{C} + A\bar{B}C$

$= \bar{A} + A\bar{B}C + B + \bar{C}$

$= \bar{A} + \bar{B}C + B + \bar{C}$

$= \bar{A} + B + \bar{C} + \bar{B}C$

$= \bar{A} + B + \bar{C} + \bar{B}$

$= \bar{A} + \bar{C} + 1$

$= 1.$

Q. $= \overline{\bar{A}B + \bar{A} + AB}$

$= \overline{\bar{A} + \bar{B} + \bar{A} + AB}$

$= \overline{\bar{A} + 1}$

$= \bar{1}$

$= 0$

Sum of Products (SOP)

$f(A,B,C) = ABC + A\bar{B}\bar{C}$

(+ : Sum; ABC, $A\bar{B}\bar{C}$: Product term.)

Product of Sum form

$f(A,B,C) = (A+B)\cdot(\bar{B}+C)$

(· : Product; $(A+B)$, $(\bar{B}+C)$: Sum term.)

Standard SOP & standard POS form

* If each term in SOP & POS forms contains all the literals then the SOP form or POS form is known as standard / canonical SOP / POS form

Converting Expressions in Standard SOP or POS form

Step-1 :- find the missing literal in each product term if any.

Step-2 : AND each product term having missing literal/s with term/s form by ORing the literal and its complement

Step-3 :- Expand the terms by applying distributive law and reorder the literals in the product term

Step-4 Reduce the expression by omitting repeated product terms. If any. as $A+A=A$.

Q. Convert the given expression in standard SOP form.

$f(A,B,C) = AC + AB + BC$

AC is missing B; AB is missing C; BC is missing A.

Q Steps to convert POS to standard POS form.

Step-1: find the missing literal in each sum term if any.

Step-2: OR each sum term having missing literals with terms form by ANDing the literal & its complement.

Step-3: Expand the terms by applying distributive law and reorder the literals in the sum term.

Step-4: Reduce the exp.

Q Convert into standard POS form

$f(A,B,C) = (A+B)(B+C)(A+C)$

Step-1 C, A, B missing.

Step-2 $(A+B)+(C\cdot\bar{C})\ (B+C)(A\cdot\bar{A})\ \ A+C+(B\cdot\bar{B})$

Step-3 $(A+B+C)\ (A+B+\bar{C})\ (B+C+A)\ (B+C+\bar{A})\ (A+C+B)$

$(A+C+\bar{B})$

repeated term.

Since:- $A+BC = (A+B)(A+C)$

Omit the repeated term

$(A+B+C)\ (A+B+\bar{C})\ (B+C+\bar{A})\ (A+C+\bar{B})$

Q. Convert the given expression in standard POS form.

$Y = A\cdot(A+B+C)$

↳ B & C are missing

$= (A+B\bar{B}+C\bar{C})\ (\bar{A}+B+C)$

$= (A+B\cdot\bar{B}+C)\ (A+B\cdot\bar{B}+\bar{C})\ (A+B+C)$

$= (A+B+C)\ (A+\bar{B}+C)\ (A+B+\bar{C})\ (A+\bar{B}+\bar{C})$

Omit → $(A+B+C)\ (A+\bar{B}+C)\ (A+B+\bar{C})\ (A+\bar{B}+\bar{C})$ $(A+B+C)$

Q $(P+\bar{Q})(P+R)$

POS form

$$f(A,B,C) = (A+\bar{B}+C)(\bar{A}+B+\bar{C})$$

$$= M_2 \cdot M_5$$

A	B	C	Y	
0	0	0	1	
0	0	1	1	
0	1	0	0	$\rightarrow A+\bar{B}+C$
0	1	1	1	
1	0	0	1	
1	0	1	0	$\rightarrow \bar{A}+B+\bar{C}$
1	1	0	1	
1	1	1	1	

Note:- The POS and SOP form derived from same Truth Table are logically equivalent. POS and SOP form are complementary to each other.

$$f(A,B,C) = m_0 + m_1 + m_3 + m_4 + m_6 + m_7$$

$$= M_2 \cdot M_5$$

if $f(A,B,C,D) = \Sigma m(0,2,4,6,8,10,12,14)$

$f(A,B,C,D) = \Pi M(1,3,5,7,9,11,13,15)$

Q. Express $F = A + B'C$ as sum of minterms.

$$A + B'C = A(B+\bar{B})(C+\bar{C}) + (A+\bar{A})\bar{B}C$$

$$= (AB + A\bar{B})(C+\bar{C}) + A\bar{B}C + \bar{A}\bar{B}C$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}C$$

$$= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + \bar{A}\bar{B}C$$

$\bar{A} \rightarrow 0$, $A \rightarrow 1$

~~$F = \Sigma m(0,1,2,3,6$~~

$$F = \Sigma m(1,4,5,6,7)$$

Way of Grouping

1. Pairs
2. Quad
3. Octet

* Top & corresponding bottom cells are adjacent
* Leftmost & Right-most cells are adjacent.

A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	0	1	1
A	0	0	0	0

$$\bar{A}BC + \bar{A}B\bar{C}$$
$$= \bar{A}B(C+\bar{C})$$
$$= \bar{A}B$$

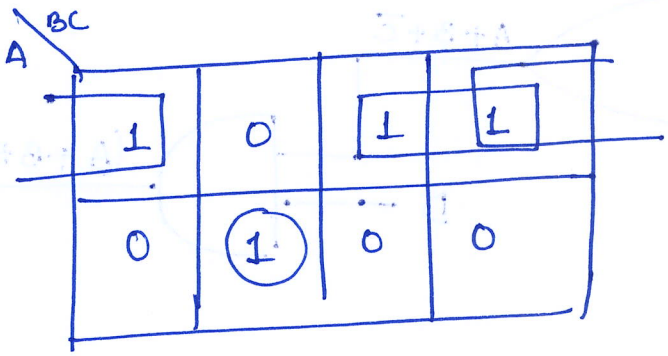
We can write directly the common terms from k-Map.
first write row followed by columns.

$$= \bar{A}B$$

A \ BC	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$	0	1	0	0
A	0	1	0	0

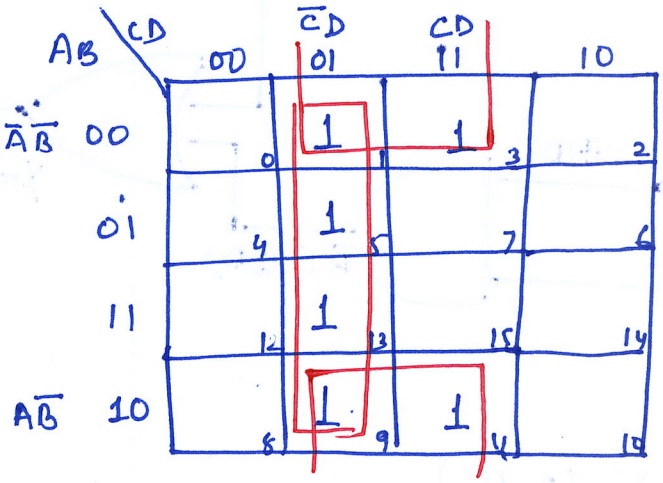
Ans. $\bar{B}C$.

Q. $Y = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C$



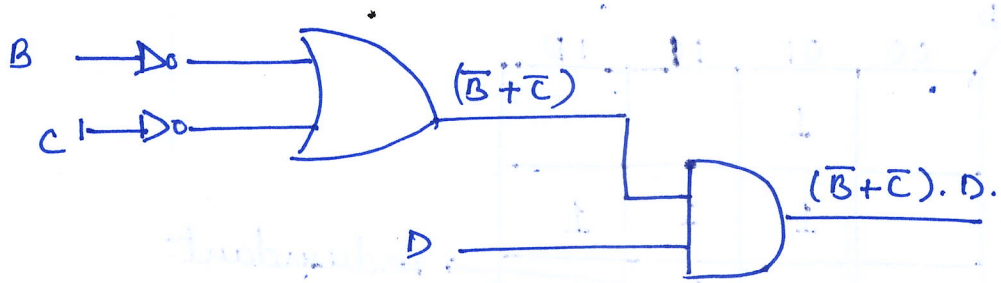
$$Y = \bar{A}B + \bar{A}C + A\bar{B}C$$

$Y = \Sigma m(1, 3, 5, 9, 11, 13)$



$Y = \bar{C}D + \bar{B}D$

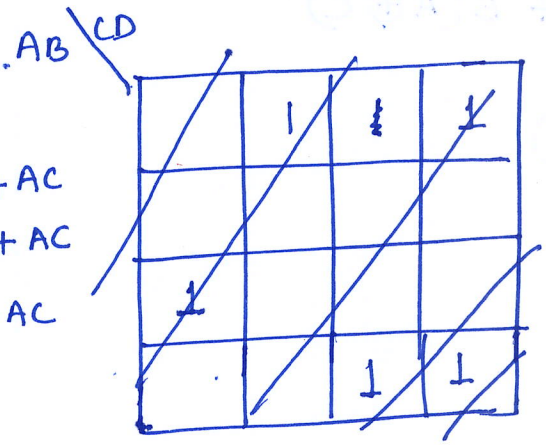
$= (\bar{B} + \bar{C})D$



Q. $Y = \Sigma m(1, 3, 4, 5, 7, 9, 11, 13, 15)$

Ans. $Y = \bar{A}B\bar{C} + D.$

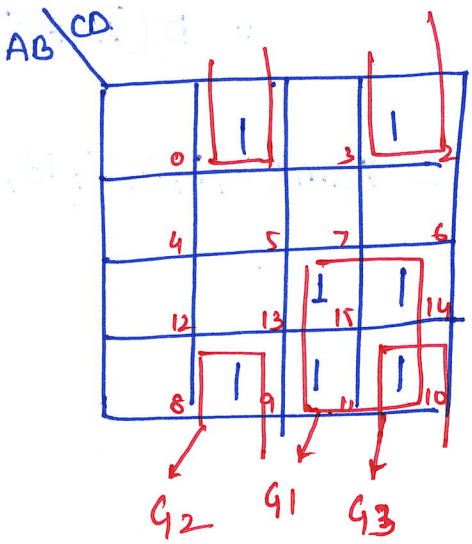
Q $Y = \Sigma m(1, 2, 9, 10, 11, 14, 15)$



$Y = \bar{B}\bar{C}D + \bar{B}C\bar{D} + AC$

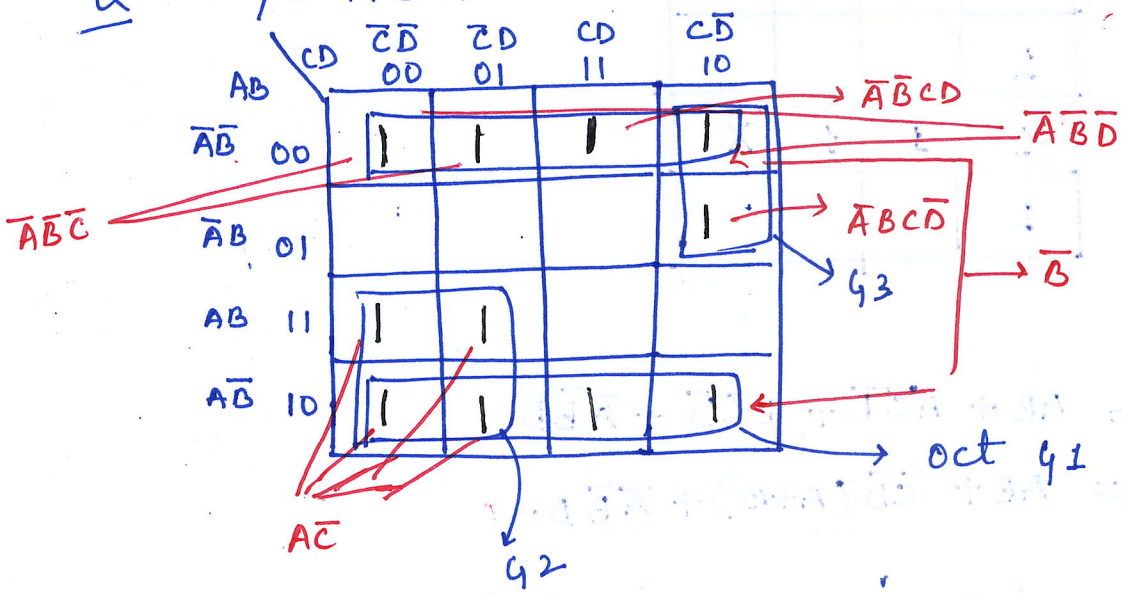
$= \bar{B}(\bar{C}D + C\bar{D}) + AC$

$= \bar{B}(C \oplus D) + AC$



Minimization of Logic function Not Specified in Standard SOP form

Q $Y = \bar{A}\bar{B}CD + \bar{A}BC\bar{D} + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{D} + A\bar{C} + \bar{B}$



$$Y = \bar{B} + A\bar{C} + \bar{A}C\bar{D}$$

Don't Care Conditions

The don't care condition (x) may be assumed to be 0 or 1 as per the need of simplification.

$$Y = \Sigma m(1,3,7,11,15) + d(0,2,5)$$

AB \ CD	00	01	11	10
00	X	1	1	X
01	0	X	1	0
11	0	0	1	0
10	0	0	1	0

$$Y = CD + \bar{A}\bar{B}$$

NAND - NAND Implementation

The SOP eq^n's are implemented using AND and OR Gates.

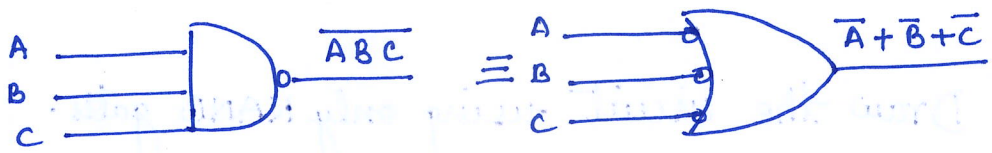
$$Y = AB\bar{C} + ABC + \bar{A}BC$$

can be implemented using AND-OR-NOT gates.

but since NAND is a universal gate and everything can be implemented using NAND.

According to Demorgan's theorem.

NAND $\equiv$ Bubbled OR.



Replace every AND $\longrightarrow$ NAND

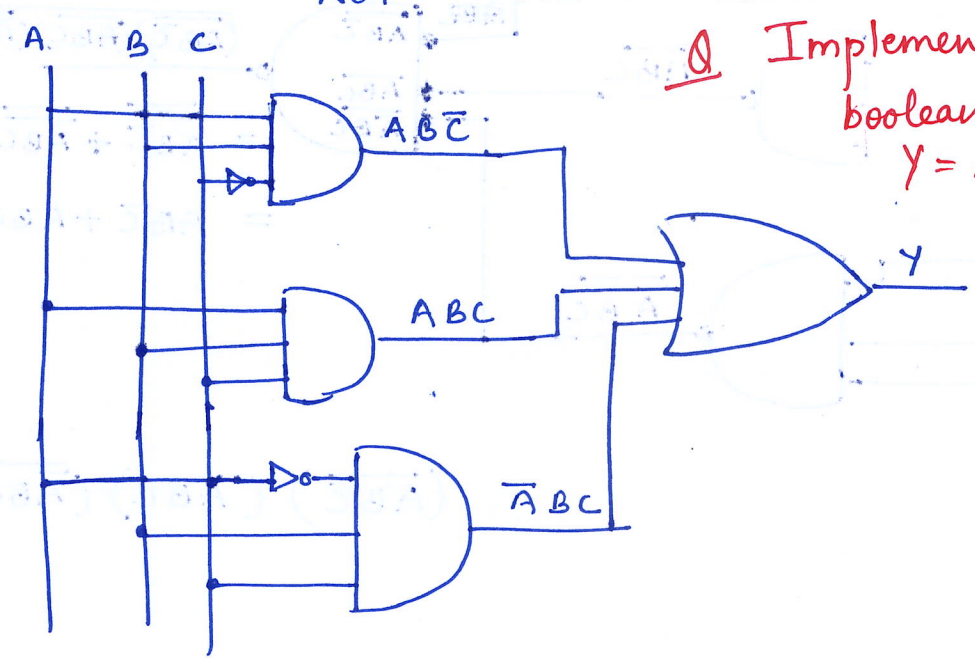
OR $\longrightarrow$ Bubbled OR

NOT $\longrightarrow$ NAND Inverter

Q Implement the following boolean

$Y = AB\bar{C} + ABC + \bar{A}BC$

using NAND gates

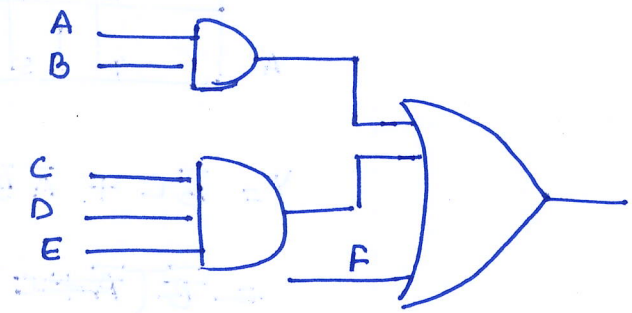


Realization using AND-OR-NOT

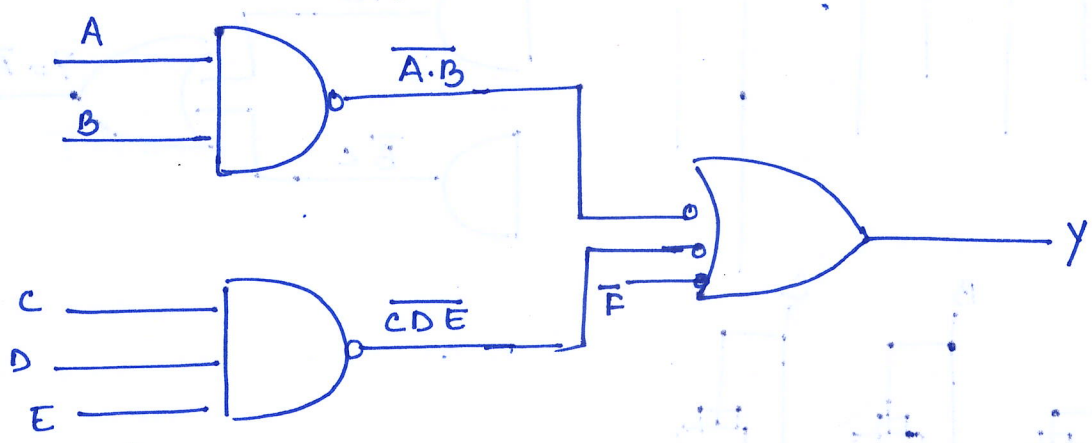
Q. Implement the following Boolean equation using only NAND gates / (NAND-NAND) logic

$$Y = AB + CDE + F$$

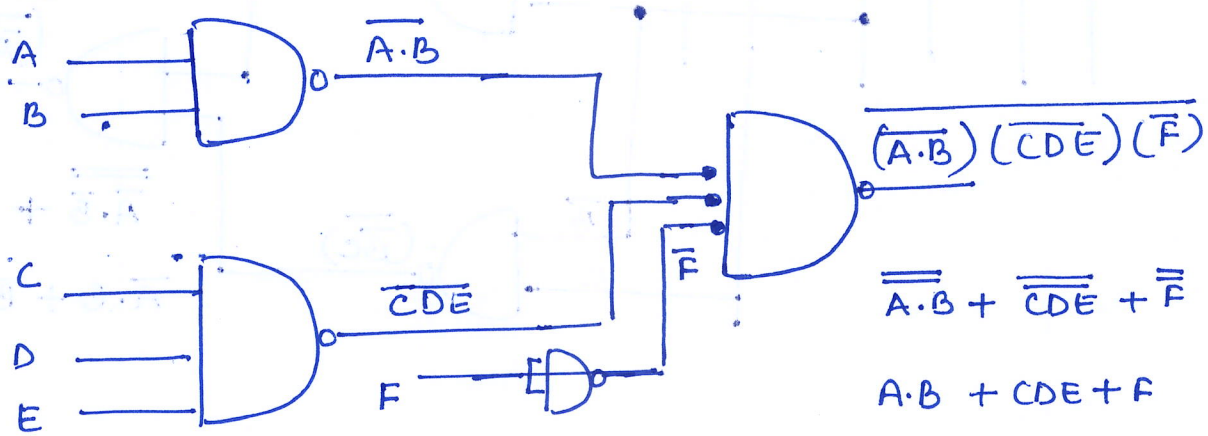
Realization using basic gates



Step-2 Replace AND → NAND
OR → Bubbled OR
NOT → NAND Inverter



Circuit using only NAND



Five Variable K-Map

The number of cells in a five variable K-Map is 32.

Two 16-cell K-maps are used as it is difficult to identify adjacent cell for grouping in 32-cell K-Map

A=0

BC \ DE	00	01	11	10
00				
01				
11				
10				

A=1

BC \ DE	00	01	11	10
00				
01				
11				
10				

→ ~~Easid~~ Consider them as superimposed on each other.

→ Adjacency in 5-variable K-Map

→ Each cell in first 16-cell K-map is adjacent to corresponding cell in the second 16-cell K-map

→ Every row and every col. in first 16-cell K-map is adjacent to corresponding rows & col. in the second 16-cell K-Map

→ The leftmost col of first K-map is not adjacent to rightmost col. of second K-map.

→ The top row of first K-map is not adjacent to ~~right~~ bottom row of the second K-map

Q. Simplify the following eqn using K-Map.

(i) $f(A,B,C,D,E) = \sum m(0,2,5,7,13,15,18,20,21,23,28,29,31)$

↳ Ans:- $Y = \bar{A}\bar{B}\bar{C}\bar{E} + \bar{B}\bar{C}D\bar{E} + CE + AC\bar{D}$

(ii) $f(A,B,C,D,E) = \sum m(0,1,2,3,10,12,13,14,20,21,22,23,26,27,28,29,30,31)$

↳ Ans $Y = \bar{A}\bar{B}\bar{C} + AC + ABD + BD\bar{E} + BC\bar{D}$

Product of Sum Method.

Two variable in POS form

A \ B	B = 0	$\bar{B}$ = 1
A = 0	A+B	$A+\bar{B}$
$\bar{A}$ = 1	$\bar{A}+B$	$\bar{A}+\bar{B}$

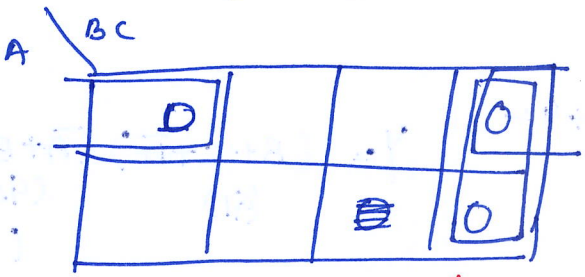
Three variable in POS form

A \ BC	00	01	11	10
0	A+B+C	$A+B+\bar{C}$	$A+\bar{B}+\bar{C}$	$A+\bar{B}+C$
1	$\bar{A}+B+C$	$\bar{A}+B+\bar{C}$	$\bar{A}+\bar{B}+\bar{C}$	$\bar{A}+\bar{B}+C$

Q. Represent the following standard POS exp. on K-Map.

$Y = (A+B+C)(A+\bar{B}+C)(\bar{A}+\bar{B}+C)$

$= M_0 \cdot M_2 \cdot M_6$



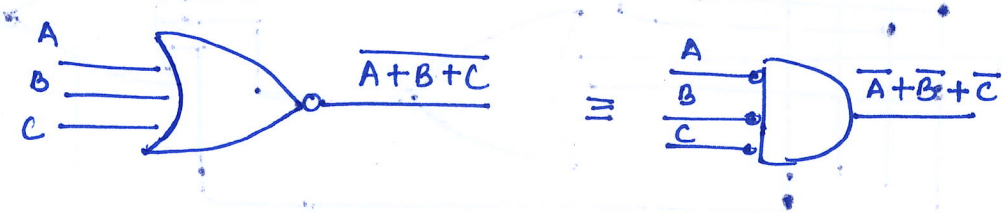
Q Represent the following standard POS.

$Y = (A+B+C+\bar{D})(A+B+\bar{C}+\bar{D})(A+\bar{B}+\bar{C}+\bar{D})(\bar{A}+\bar{B}+C+D)$

The NOR-NOR Implementation

According to Demorgan's law

NOR ≡ Bubbled AND

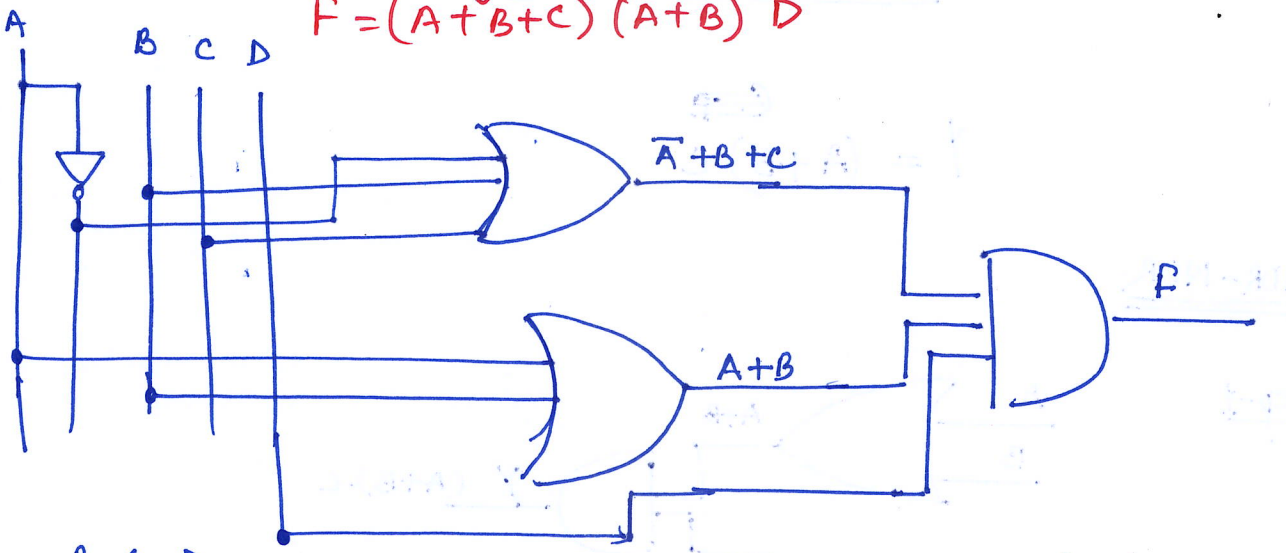


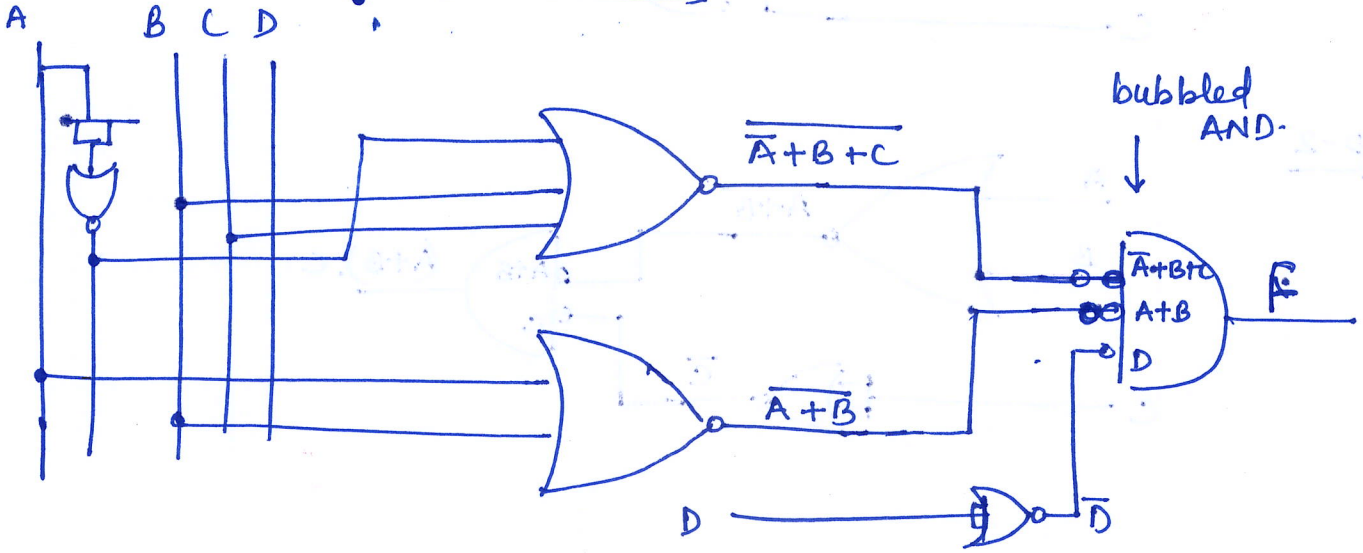
AND-OR-NOT logic to NOR-NOR

* Draw AND-OR-NOT realization.
* Replace OR → NOR
  AND → Bubbled AND
  NOT → NOR Inverter.

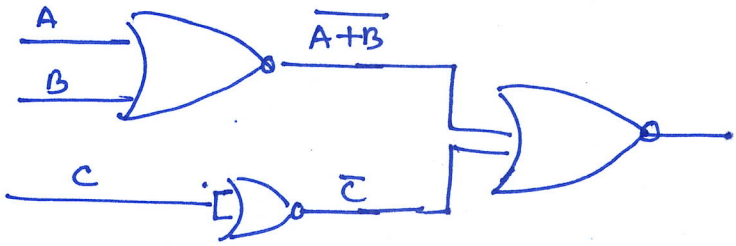
Q Implement the following Boolean function using only NOR Gates.

$F = (\overline{A}+B+C)(A+B)D$





Step-3



QUINE – McCLUSKEY METHOD